Qlassifier: a product-state quantum kernel classifier with zero-circuit tuning, benchmarked against quantum and classical baselines

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Off-the-shelf quantum kernel classifiers often fail on real data: on the Breast Cancer data, the standard ZZ-feature-map quantum support vector classifier reaches an accuracy of 0.677, barely above the 0.626 of always predicting the majority class. We present Qlassifier, a quantum kernel classifier built from single-qubit product states. It encodes two standardised features per qubit, reads three Pauli settings per sample, applies a Gaussian kernel to the measured Bloch vectors and tunes bandwidth and margin by cross-validation on exact features, at zero circuit cost. An opt-in geometry mode gives each periodic column its own qubit circle, which turns the kernel into a seam-free product of von Mises kernels. We compare Qlassifier with the standard classical and quantum baselines and four published quantum kernels, on five real datasets, under one protocol: same splits, same features, equal tuning budgets, 10 seeds and corrected resampled t statistics. Qlassifier beats the standard quantum baseline by 22.6 to 37.6 pp BA on four UCI datasets, on every seed, and has the highest mean BA over these datasets of all quantum models in the harness. It is statistically tied with the best tuned classical model on every dataset. Under three IBM device noise models it loses on average at most 0.19 pp accuracy (worst seed 0.94 pp). A first run on the IBM Heron processor ibm_fez (5 seeds, 410 QPU seconds) gives 0.875–0.975 accuracy on 40-row test subsets, against 0.874–0.975 predicted by the device model. Of three further UCI datasets, Qlassifier beats the standard baseline on two and ties it on the third. On Breast Cancer it needs 4 qubits for 8 features, no two-qubit gates and 1,420 times fewer circuits than a bandwidth-tuned fidelity-kernel pipeline (86 times fewer than its final Gram matrices alone). Product states are classically simulable; we make no claim of quantum advantage and report the exact classical twin of the geometry mode alongside it.

I. INTRODUCTION

Quantum kernel methods are the most common entry point to quantum machine learning on real data. The standard recipe is a support vector classifier on the fidelity kernel of the ZZ feature map [1], as shipped by Qiskit [2]. On real data this recipe often collapses. In our harness it reaches a mean BA of 0.571 on Breast Cancer, 0.698 on Ionosphere, 0.545 on Sonar and 0.533 on Heart, where 0.5 is chance (Fig. 1).

The known fixes are expensive. Tuning the input scale (the "bandwidth") makes fidelity kernels work [3–5], but on a device each candidate bandwidth needs a new Gram matrix, with one circuit per pair of samples. Training the kernel itself [6, 7] needs a new Gram matrix at every optimisation step. Deep entangled feature maps also lose most of their signal on today's hardware: the pinned ZZ circuit needs 393 to 428 two-qubit gates on 8 qubits after transpilation to IBM devices (Table IX). Finally, well-tuned quantum kernels tend to behave like classical RBF kernels [8, 9], and public tabular data rarely leaves room for a quantum edge [10].

Qlassifier keeps its circuits fixed and tunes only after the measurement. This paper reports:

- a product-state kernel classifier with zero-circuit self-tuning and a three-setting hardware protocol (Sec. II);
- a geometry mode for angle-valued data, with its exact classical twin (Sec. II D);
- a benchmark against the standard classical and quantum baselines and four published quantum kernels, included in the library (Sec. III);
- results on accuracy, noise robustness, cost and run time, with confidence intervals (Sec. IV), and the limitations (Sec. VI);
- a run on an IBM Heron processor, compared with a prediction written before submission (Sec. IV G), a noise sweep to 3 times the device error rates (Sec. IV F) and three further datasets (Sec. IV H).

The library is open source (MIT licence) at github.com/szymon6/qlassifier. All numbers in this paper are generated from the saved benchmark results by a script; Sec. VII shows how to rerun them.

II. METHOD

A. Core encoding

Each plain feature is standardised with the mean and standard deviation of the training rows, giving $\tilde{x}$. Qubit q carries the pair $(\tilde{x}_{2q}, \tilde{x}_{2q+1})$:

$$|\psi_q(\mathbf{x})\rangle = R_Y(c_0\tilde{x}_{2q+1})\, R_X(c_0\tilde{x}_{2q})\,|0\rangle, \qquad (1)$$

so d features need $n = \lceil d/2 \rceil$ qubits and no two-qubit gate. The angle scale $c_0 = 2/\sqrt{\max(d, 8)}$ is fixed, not tuned. It keeps standardised values inside one period of the rotation (the cap at 8 avoids folding at small d). With

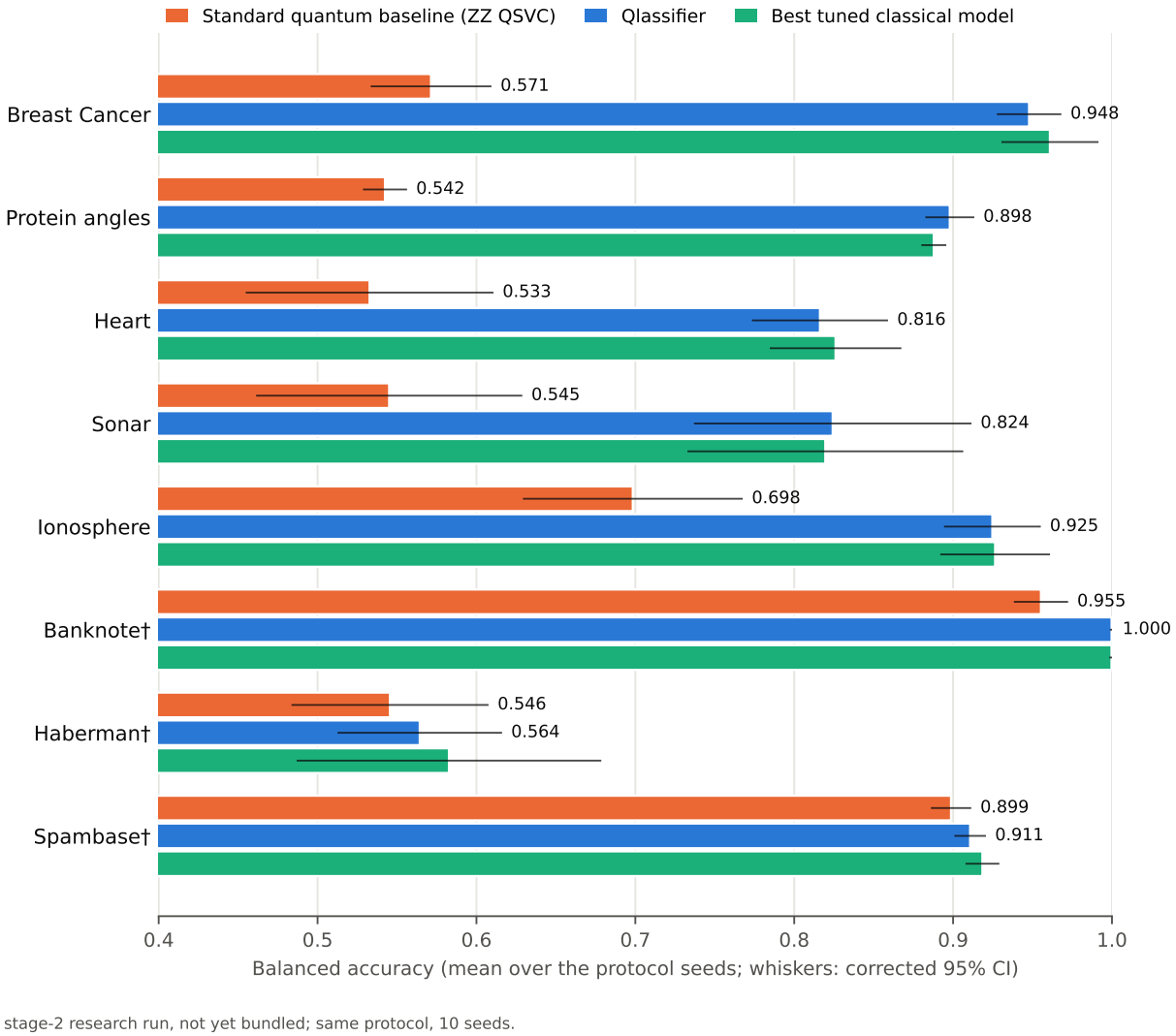


FIG. 1. Balanced accuracy of Qlassifier against the standard quantum baseline (ZZ feature map QSVC) and the best tuned classical model on every dataset of this paper. Means over the protocol's seeds, whiskers are corrected confidence intervals. Datasets are sorted by Qlassifier's gain over the standard baseline. Spambase, Banknote and Haberman are the additional datasets of Sec. IV H. Script: `scripts/wins_chart.py`.

$a = c_0\tilde{x}_{2q}$ and $b = c_0\tilde{x}_{2q+1}$, the Bloch vector of qubit q is

$$\mathbf{r}_q = (\cos a \sin b,\ -\sin a,\ \cos a \cos b). \tag{2}$$

This dense angle encoding is known [11, 12]; what follows is how we read it out and tune it.

B. Readout and kernel

Three circuits per sample measure all qubits in the X, Y and Z bases, which gives $F(\mathbf{x}) = (\mathbf{r}_1, \ldots, \mathbf{r}_n) \in \mathbb{R}^{3n}$. The kernel is Gaussian on these features:

$$k(\mathbf{x}, \mathbf{x}') = e^{-\gamma \|F(\mathbf{x}) - F(\mathbf{x}')\|^2} = \prod_{q=1}^{n} e^{-4\gamma\,[1 - f_q(\mathbf{x}, \mathbf{x}')]}, \tag{3}$$

where $f_q = (1 + \mathbf{r}_q \cdot \mathbf{r}'_q)/2$ is the single-qubit fidelity, $\gamma = g\,\gamma_0$ and $\gamma_0 = 1/\sum_k \mathrm{Var}(F_k)$ over the training rows. This is a projected quantum kernel [13] on a product state, or equivalently a tempered fidelity kernel. A support vector machine uses the precomputed Gram matrix. For small angles $\|F(\mathbf{x}) - F(\mathbf{x}')\|^2 \approx c_0^2 \|\tilde{\mathbf{x}} - \tilde{\mathbf{x}}'\|^2$, so the core kernel is close to an RBF kernel. This explains the ties with RBF-SVM below.

C. Self-tuning at zero circuits

The two hyperparameters act after the measurement: g scales the kernel width and C the margin. Qlassifier therefore tunes them on exact (closed-form) features and never runs a circuit to tune. It uses 5-fold stratified cross-

validation on the training rows, scored by balanced accuracy, over a ridge-following grid [14]: $g \in \{4^{-3}, \ldots, 4^{2}\}$ and $C = \tilde{C}/g$ with $\tilde{C} \in \{10^{-1}, \ldots, 10^{3}\}$, 30 configurations in total. Ties go to the first configuration in grid order. A device measures only the final features of the training and test rows. Probabilities come from a Platt sigmoid fitted on out-of-fold decision values.

D. Geometry mode for angle-valued data

Some columns are angles: dihedral angles, wind directions, times of day, longitudes. A periodic column x with period P gets its own qubit, $|\psi(x)\rangle = R_Y(t)|0\rangle$ with $t = 2\pi x/P$, so $\mathbf{r} = (\sin t, 0, \cos t)$. The kernel factor of that qubit is

$$e^{-\gamma\|\mathbf{r}-\mathbf{r}'\|^2} = e^{-2\gamma\,[1-\cos(t-t')]}, \tag{4}$$

a von Mises kernel with concentration 2γ [15, 16]. It has no seam: 359° and 1° are neighbours. A (latitude, longitude) pair maps to one qubit with $R_Z(\varphi)R_Y(\theta)$, $\theta = 90° - \text{latitude}$, $\varphi = \text{longitude}$; its Bloch vector is the direction itself and its kernel factor is a von Mises–Fisher kernel on the sphere.

Two circles cannot share a qubit. The map of Eq. (1) sends (a, b) and $(\pi - a, b + \pi)$ to the same Bloch vector, because the torus does not embed in the sphere. Geometry mode therefore uses one qubit per circle, more than the $\lceil d/2 \rceil$ of the core encoding, and our hardware target counts it by this rule (Sec. VI). It is still free of two-qubit gates.

Classical twin. When every column is angular, the geometry-mode Gram matrix equals that of an RBF kernel on hand-made features: $(\sin t, \cos t)$ per circle and the unit vector per direction. Both give the same squared distances, and the zero-variance Y components leave γ_0 unchanged. The library tests this identity to 10^{-12}. Any gain of geometry mode therefore comes from the geometric prior, which a classical model can also use through feature engineering. We report the twin next to every geometry-mode result.

E. Execution on devices

The `backend` parameter selects where Bloch vectors come from: the closed form (`"exact"`, the default), binomial shot sampling of each Pauli expectation (`"shots"`, the exact law for product states) or the noise models of IBM devices (`"fake_torino"`, `"fake_fez"`, `"fake_brisbane"`: Heron r1, Heron r2 and Eagle r3, simulated with Qiskit Aer). Circuits are transpiled once at optimisation level 3; the X and Y circuits reuse the physical qubits chosen for the Z circuit. Each component is estimated as $1 - 2P(1)$ from 1,024 shots, and each measured Bloch vector is renormalised to unit length, a per-qubit mitigation that costs no extra circuit. The circuits are product states, and the device noise models act qubit by qubit: single-qubit gate errors and readout errors, no crosstalk, no idle noise. So the library simulates each qubit alone, on the physical qubit that the full-register transpilation chose, at a cost linear in the number of qubits. This reproduces the full-register marginals exactly: over 450 marginals the largest difference was 3.3×10^{-16} (experiment e14 in the repository). The library checks these conditions at run time and otherwise simulates the full register.

F. Resources

Per circuit, the core encoding needs $\lceil d/2 \rceil$ qubits (4 for $d = 8$), no two-qubit gate and a transpiled depth of 6 on the three devices. Training and prediction need 3 circuits per sample: 1,707 circuits for the 569 Breast Cancer rows. Tuning needs none. Classically, the Gram matrix costs $O(N^2 n)$ time and N^2 memory.

III. BENCHMARK PROTOCOL

The benchmark follows the fair-comparison rules of the Qiskit Fall Fest 2026 QML challenge and was fixed before the method search (`experiments/classifier/README.md`, Sec. 2, in the repository). It ships inside the library: `qlassifier benchmark` reruns it.

Data. Table I lists the five bundled datasets: four UCI datasets [17–20] and protein backbone dihedral angles from the Protein Data Bank [21]. The protein task asks whether residue i lies in a β-strand, from the (ϕ, ψ) angles of residues $i - 2$ to $i + 2$. The positive class is always the minority class.

Splits. Seeds 0 to 9: one stratified 30% test split per seed (protein: grouped by PDB chain), identical for every model. All preprocessing is fitted on training rows only.

Models. Majority class; logistic regression, RBF-SVM, random forest, histogram gradient boosting and an MLP (scikit-learn [22]); the pinned standard quantum baseline, a QSVC on the fidelity kernel of the ZZ feature map (2 repetitions, full entanglement, inputs scaled to $[0, \pi]$); four published quantum kernels re-implemented under the same protocol: the bandwidth-tuned fidelity kernel with full and linear entanglement [3, 4], the projected quantum kernel [13] and the separable kernel [9]; Qlassifier; and on the protein data the classical twin as a control. Exact simulation is used for every quantum model in the leaderboard.

Tuning. Every tuned model gets 30 configurations, scored by 5-fold cross-validation on balanced accuracy inside the training split, then refits on the whole training split. The pinned ZZ baseline keeps its 4 values of C. No test label is seen before the final evaluation.

Statistics. Repeated random splits overlap, so a plain t-test over seeds is anti-conservative. Every interval is a

TABLE I. Bundled datasets. UCI sets: every model gets the same 8 PCA features, fitted inside every training fold. Protein: every model gets the same 10 raw angles; geometry mode declares them as circles of period 360°. Chain-grouped splits keep every protein chain on one side of each split.

Dataset	Rows	Features	Positive class	Features used	Split	Licence
Breast Cancer	569	30	malignant (212)	PCA to 8	stratified	CC BY 4.0 (UCI); BSD-3-Clause (scikit-learn copy)
Ionosphere	351	33	bad (126)	PCA to 8	stratified	CC BY 4.0
Sonar	208	60	rock (97)	PCA to 8	stratified	CC BY 4.0
Heart	297	13	disease (137)	PCA to 8	stratified	CC BY 4.0
Protein angles	4,000	10	strand (996)	10 angles (raw)	chain-grouped	CC0 1.0

corrected resampled t interval [23],

$$\mathrm{SE} = s\sqrt{1/J + n_{\mathrm{test}}/n_{\mathrm{train}}}, \tag{5}$$

with J seeds and $J-1$ degrees of freedom. Paired differences (Qlassifier minus a model) use the same correction. A model is "tied with the best" when the 95% interval of its paired difference to the top model contains 0. The best classical model is chosen after the fact, which is conservative for a quantum claim. The paired intervals (one per model and dataset) are reported without a multiple-comparison correction: they describe ties, not discoveries, and each target rests on one pre-registered test (the earlier protein tests used Bonferroni-corrected intervals, Table IV). Reading a tie from an uncorrected interval is conservative for a quantum win.

Grid edges. Equal effort is counted in configurations, not seconds. The chosen configuration sits on an edge of the grid for at least half the seeds for Random forest 1.0 (Breast Cancer, Ionosphere); FQK-bw full 0.9 (Breast Cancer); PQK 0.9 (Sonar, Protein angles); FQK-bw linear 0.8 (Heart); Gradient boosting 0.8 (Breast Cancer); Qlassifier 0.6 (Heart); Separable 0.6 (Breast Cancer); RBF-SVM 0.6 (Heart) (share of seeds, datasets); Qlassifier's largest share is 0.6 (Heart). The budget is equal, but a wider grid could move these models, in either direction; the per-seed edge flag is in every `runs.csv`.

Noise study. Part A runs Qlassifier on every dataset (all rows, seeds 0 to 4) on the shot sampler and the three device models. Part B compares Qlassifier with the bandwidth-tuned fidelity kernel (linear), the projected kernel and the separable kernel on identical Breast Cancer subsets (60 training and 40 test rows, 3 seeds), because fidelity kernels need one circuit per pair of rows. Hyperparameters come from exact-mode tuning; nothing is re-tuned on noisy data.

Cost study. Each quantum model's circuit is transpiled to the three devices (median of 3 transpiler seeds). Circuit counts cover training and prediction at the chosen configuration, plus, for models whose hyperparameters change the circuit, the on-device bandwidth search.

Targets. Ten targets, BT-1 to BT-10, were fixed before the benchmark ran. The library evaluates them from the saved results (Table X).

IV. RESULTS

A. Against the standard quantum baseline

Qlassifier beats the pinned ZZ QSVC on every dataset and every seed (Table III): by 22.6 to 37.6 pp BA on the four UCI datasets and by +35.5 pp on the protein angles. On Breast Cancer the accuracy rises from 0.677 to 0.956 (BT-1). The ZZ baseline is not broken by a bug. Its input scale is fixed, so on 8 qubits its kernel matrix is close to the identity on real data, and tuning C cannot repair that [3, 4].

B. Against other quantum models

Averaged over the four UCI datasets, Qlassifier has the highest mean BA of the six quantum models in the harness (0.878; next: Separable, 0.870; BT-5). The differences between the good quantum kernels are small and mostly inside the CIs.

C. Against tuned classical models

On every dataset Qlassifier is statistically tied with the best tuned classical model (Table III). The mean differences range from −1.3 to +1.0 pp. With 10 seeds and test sets of 63 to 1,201 rows, the corrected 95% half-widths of these differences are 1.2 to 9.0 pp, so differences of about one point cannot be resolved. Ties are the expected outcome on generic tabular data, as the small-angle limit of Eq. (3) predicts.

D. Geometry mode on protein angles

On the protein angles, geometry mode ranks 1st of 13 by mean BA (0.898 ± 0.009). Its difference to the best tuned classical model, Gradient boosting, is +1.0 pp (95% CI [−0.2, +2.2], 10/10 seeds): a statistical tie (the CI contains 0). Against RBF-SVM the difference is +2.1 pp (CI [+0.8, +3.5]), against logistic regression +6.0 pp (CI [+3.9, +8.1]). Its classical twin scores 0.898: identical on all 10 seeds, as the Gram identity requires.

TABLE II. Balanced accuracy, mean ± standard deviation over 10 seeds. Bold: top mean per dataset. ≈: statistically tied with the top model (corrected 95% CI of the paired difference contains 0). The classical twin exists only for the geometry-mode dataset.

Model	Breast Cancer	Ionosphere	Sonar	Heart	Protein angles
Qlassifier	0.948 ± 0.012≈	0.925 ± 0.018≈	**0.824 ± 0.053**≈	0.816 ± 0.026≈	**0.898 ± 0.009**≈
Classical twin	—	—	—	—	**0.898 ± 0.009**≈
ZZ QSVC	0.571 ± 0.023	0.698 ± 0.042	0.545 ± 0.051	0.533 ± 0.047	0.542 ± 0.008
FQK-bw full	0.944 ± 0.018≈	0.891 ± 0.031≈	0.800 ± 0.036≈	0.823 ± 0.029≈	0.878 ± 0.013
FQK-bw linear	0.946 ± 0.020≈	0.904 ± 0.031≈	0.772 ± 0.037≈	0.800 ± 0.053≈	0.877 ± 0.011
PQK	0.950 ± 0.014≈	0.900 ± 0.035≈	0.806 ± 0.043≈	0.793 ± 0.043≈	0.878 ± 0.009
Separable	0.952 ± 0.015≈	0.898 ± 0.031≈	0.805 ± 0.040≈	**0.826 ± 0.025**≈	0.888 ± 0.008≈
Logistic regression	0.954 ± 0.018≈	0.845 ± 0.029	0.721 ± 0.046	**0.826 ± 0.025**≈	0.838 ± 0.010
RBF-SVM	0.949 ± 0.012≈	**0.926 ± 0.021**≈	0.814 ± 0.055≈	0.820 ± 0.025≈	0.877 ± 0.011
Random forest	0.923 ± 0.021	0.905 ± 0.026≈	0.784 ± 0.072≈	0.792 ± 0.030≈	0.883 ± 0.007
Gradient boosting	0.948 ± 0.018≈	0.886 ± 0.024≈	0.754 ± 0.040≈	0.784 ± 0.031≈	0.888 ± 0.005≈
MLP	**0.961 ± 0.019**≈	0.915 ± 0.030≈	0.820 ± 0.052≈	0.820 ± 0.027≈	0.884 ± 0.008≈
Majority class	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000

TABLE III. Qlassifier against the standard quantum baseline and the best tuned classical model. Δ = Qlassifier minus the other model, in percentage points (pp), with corrected 95% CIs; wins = seeds on which Qlassifier is higher.

Dataset	Qlassifier BA mean [95% CI]	Δ vs ZZ QSVC pp [95% CI]	wins	Δ vs best classical model	pp [95% CI] (wins)
Breast Cancer	0.948 [0.928, 0.968]	+37.6 [+34.6, +40.7]	10/10	MLP	−1.3 [−3.1, +0.5] (0/10)
Ionosphere	0.925 [0.894, 0.955]	+22.6 [+13.9, +31.4]	10/10	RBF-SVM	−0.2 [−3.1, +2.7] (5/10)
Sonar	0.824 [0.737, 0.912]	+27.9 [+14.7, +41.2]	10/10	MLP	+0.5 [−8.5, +9.5] (6/10)
Heart	0.816 [0.773, 0.859]	+28.3 [+20.4, +36.3]	10/10	Logistic regression	−1.0 [−3.1, +1.2] (2/10)
Protein angles	0.898 [0.883, 0.913]	+35.5 [+34.6, +36.5]	10/10	Gradient boosting	+1.0 [−0.2, +2.2] (10/10)

The protein data was chosen in an earlier search, so these seeds are a confirmation run; Table IV lists every look at this question, including the leaky residue-level split that first suggested a significant win.

E. Noise

Table V gives the accuracy lost from exact simulation to each noise model (positive: lost; negative: gained). Over the five datasets and three device models, Qlassifier's mean change lies between −1.33 and 0.19 pp, and its kernel keeps a Pearson correlation of at least 0.9968 with the exact kernel. On the paired Part B subsets, at 1,024 shots, the bandwidth-tuned fidelity kernel loses 17.5 to 22.5 pp, the separable kernel up to 14.2 pp, and Qlassifier at most 0.8 pp (BT-9). Most of the fidelity kernel's loss is sampling noise, not gate error: with device noise switched off it already loses 19.2 pp (Sec. IV F). These are means over seeds; the worst single seed lost 0.94 pp in Part A (Ionosphere) and 2.5 pp in Part B. Fig. 4 shows Breast Cancer.

On Breast Cancer the three device models give the same mean loss, +0.12 / +0.12 / +0.12 pp: with 1,024 shots and a shared sampler seed, their estimates coincide to the flip, so device differences on this data lie below the shot resolution. An earlier run of the same study, which simulated the whole register, measured +0.12 / +0.23 / +0.47 pp (e05 in the repository). The shipped run samples each qubit alone and draws a different random stream from the simulator, so the two differ by one shot-noise resampling; both stay below 0.47 pp.

F. Noise scaled beyond the device

We scaled every error channel of the FakeFez model (gate and readout errors multiplied by s, T_1 and T_2 divided by s) for $s \in \{0, 0.5, 1, 1.5, 2, 3\}$ and reran Breast Cancer at 1,024 shots. Qlassifier's mean accuracy loss is +0.12 pp at $s = 1$ and +0.12 pp at $s = 3$ (worst seed 0.58 pp; 5 seeds, all rows), and its kernel keeps $r = 0.998$ with the exact kernel at $s = 3$. On the paired subsets (3 seeds) the mean accuracy over the sweep is 0.900 for Qlassifier, 0.738 for the bandwidth-tuned fidelity kernel and 0.625 for the ZZ baseline (2 seeds at $s \in \{0, 1, 3\}$), Fig. 5. The fidelity kernel loses 19.2 pp already at $s = 0$, from 1,024-shot sampling alone, and 21.7 pp at $s = 3$: its noise loss is mostly shot noise, the price of estimating one fidelity per pair of samples. The ZZ classifier stays at the majority class throughout while its kernel correlation with the exact kernel falls from 0.92 to 0.02. Qlassifier's curve is flat because, for depth-10 single-qubit circuits, device noise lies far below the shot resolution. A scaled

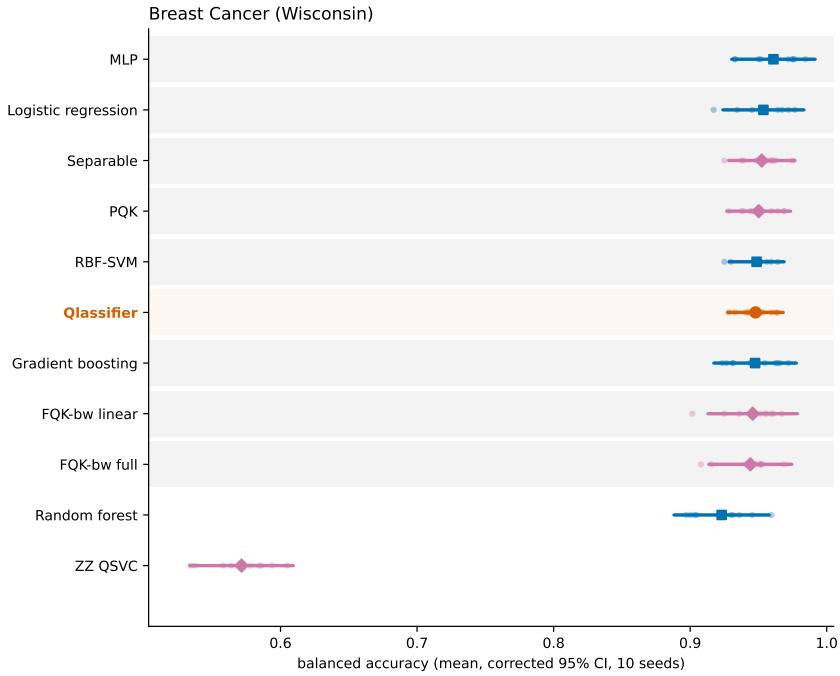


FIG. 2. Breast Cancer leaderboard: per-seed BA (dots), corrected 95% CI (bar) and mean (marker) for every model. Qlassifier (vermillion) sits inside the band tied with the best, far above the ZZ QSVC.

TABLE IV. The tuned-classical target (BT-6) on protein data, every run. Geometry mode minus the best classical model, in pp. e07: residue-level splits, which share protein chains between training and test. e11, e12: chain-grouped, pre-registered; e12 used Bonferroni-corrected 97.5% CIs for its two hypotheses. This paper: the run reported here.

Run	Data, split	Seeds	Best classical	Qlassifier	Δ BA, pp [CI]	CI level
e07	protein_strands, residue-level split	200–209	MLP 0.870	0.884	+1.4 [+0.3, +2.4]	95%
e11	protein_strands, chain-grouped	200–209	MLP 0.872	0.879	+0.7 [−0.6, +2.1]	95%
e12	protein_div6, chain-grouped	300–309	MLP 0.887	0.889	+0.25 [−1.13, +1.62]	97.5%
e12	protein_div10, chain-grouped	300–309	Gradient boosting 0.889	0.897	+0.81 [−0.61, +2.23]	97.5%
this paper	protein_angles, chain-grouped	0–9	Gradient boosting 0.888	0.898	+1.0 [−0.2, +2.2]	95%

noise model does not replace a hardware run (Sec. IV G).

G. On a real quantum processor

On 2026-10-25 we ran Qlassifier on ibm_fez, an IBM Heron r2 processor, through Qiskit Runtime (job `dbesrqnpc3tc739mb92g`, 09:37–09:39 UTC). The task is the Part B Breast Cancer subset of seed 0 (60 training and 40 test rows). Training, tuning and the exact prediction used the simulator; the device measured the Bloch readout of all 60 + 40 rows: 300 circuits (three Pauli settings per row) at 1,024 shots, 307,200 shots in all. The circuits use 4 physical qubits (4, 2, 130, 50), depth 10 and 0 two-qubit gates after transpilation; Table VII gives their calibration at submission. The job was billed 82 QPU seconds (0.273 s per circuit), inside the 79–110 s predicted by our QPU cost model, which brackets the published per-circuit timings of Table IX's sources as well.

Before submission we wrote down the FakeFez model's prediction for this exact subset: accuracy 0.880 ± 0.010 over 20 shot-noise draws (range 0.875–0.900; exact simulation 0.875), and the pass rule, within 2 pp of the prediction. The hardware gave accuracy 0.900 (36 of 40 rows; balanced accuracy 0.907), +2.0 pp above the prediction and at the top of its shot-noise range. The measured kernel correlates with the exact one at $r = 0.999$ (training block 0.998), and the mean Bloch-vector error is 0.020 against 0.017 simulated. The result is consistent with the prediction. We then ran 4 more seeds of the same design on the same device (Table VI); their predictions were computed by the same procedure after the runs, so they are checks, not pre-registered tests. Over all 5 seeds the hardware accuracy is 0.875–0.975 (mean 0.935) against predictions of 0.874–0.975 (mean 0.932); 5 of 5 seeds lie within 2 pp of their prediction and 5 inside the shot-noise range; the mean deviation is +0.3 pp (largest 2.0 pp), the kernel correlation at least 0.998, and the whole study cost 410 QPU seconds. The evidence is limited to a single de-

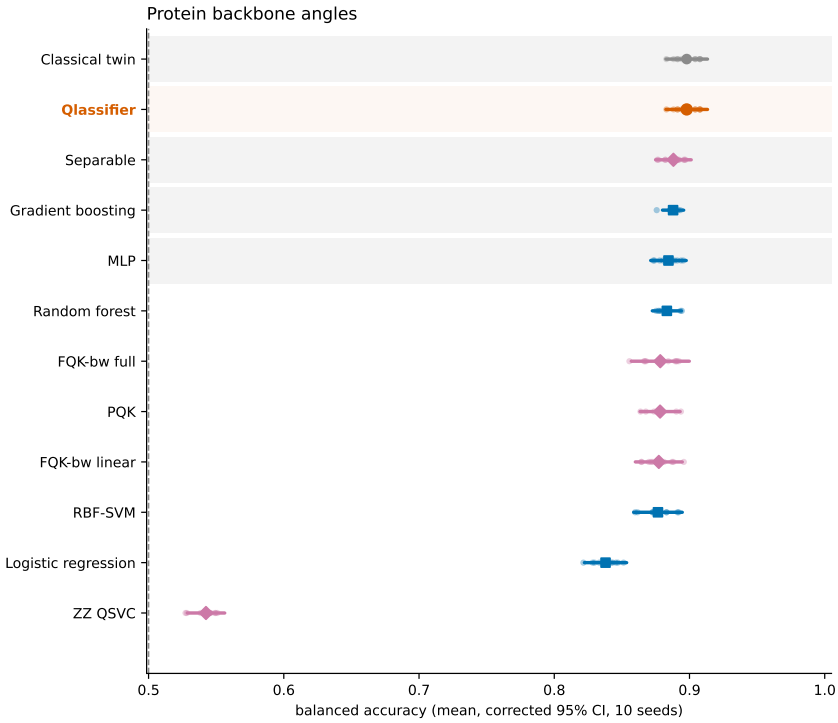


FIG. 3. Protein backbone angles (chain-grouped splits): leaderboard as in Fig. 2. The classical twin (grey) computes the same Gram matrix as geometry mode from hand-made $(\sin, \cos)$ features.

TABLE V. Accuracy lost under noise, in pp: mean over seeds (worst seed in brackets); 1,024 shots. Part A: Qlassifier on all rows, 5 seeds. Part B: four quantum models on identical Breast Cancer subsets. Kernel r: lowest mean Pearson correlation of the noisy and exact kernels over the noise models.

Part	Model, dataset	shots	Torino	Fez	Brisbane	kernel r (min)
A	Qlassifier, Breast Cancer	−0.12 (+0.6)	+0.12 (+0.6)	+0.12 (+0.6)	+0.12 (+0.6)	0.9983
A	Qlassifier, Ionosphere	−0.19 (0.0)	−0.38 (+0.9)	−0.75 (+0.9)	−0.38 (+0.9)	0.9973
A	Qlassifier, Sonar	−0.32 (+1.6)	−0.32 (0.0)	−0.63 (0.0)	−0.63 (0.0)	0.9968
A	Qlassifier, Heart	−1.56 (0.0)	−0.89 (0.0)	−1.33 (0.0)	−1.33 (0.0)	0.9980
A	Qlassifier, Protein angles	+0.18 (+0.4)	+0.17 (+0.4)	+0.17 (+0.4)	+0.19 (+0.3)	0.9998
B	Qlassifier, Breast Cancer 60/40	—	+0.8 (+2.5)	+0.8 (+2.5)	+0.8 (+2.5)	0.9981
B	FQK-bw linear, Breast Cancer 60/40	—	+22.5 (+27.5)	+20.0 (+27.5)	+17.5 (+35.0)	0.9069
B	PQK, Breast Cancer 60/40	—	0.0 (+2.5)	+1.7 (+7.5)	0.0 (0.0)	0.9821
B	Separable, Breast Cancer 60/40	—	+14.2 (+30.0)	+13.3 (+30.0)	+10.8 (+22.5)	0.8673

vice and calibration day, with 40 test rows per seed, so one row is 2.5 pp and the pass rule is a one-row tolerance. The full records (job ids, timestamps, transpiled circuits, raw counts, calibration before and after) are in the repository (Sec. VII).

H. Three more datasets

We ran the same protocol (same splits, features, tuning budgets and 10 seeds) on three more UCI datasets (Table VIII). On Spambase (4,601 rows) Qlassifier reaches BA 0.911 against 0.899 for the ZZ baseline (+1.2 pp, CI [+0.8, +1.6], 10/10 seeds); on Banknote 1.000 against 0.955 (+4.4 pp). Both tie the best classical model (−0.8 pp vs Gradient boosting, CI [−2.0, +0.4]; 0.0 pp vs MLP), but on Spambase the MLP is ahead of Qlassifier by a small, significant margin (−0.7 pp, CI [−1.2, −0.2]): a tie with the best model does not mean a tie with every model. On Haberman (306 rows) every model lands between 0.564 and 0.583, and every pair of models ties. The ZZ margin is dataset-specific: +1.2 and +4.4 pp here against 22.6 to 37.6 pp on the first four datasets. These datasets are not bundled in the released library yet; the results come from the research scripts in the repository.

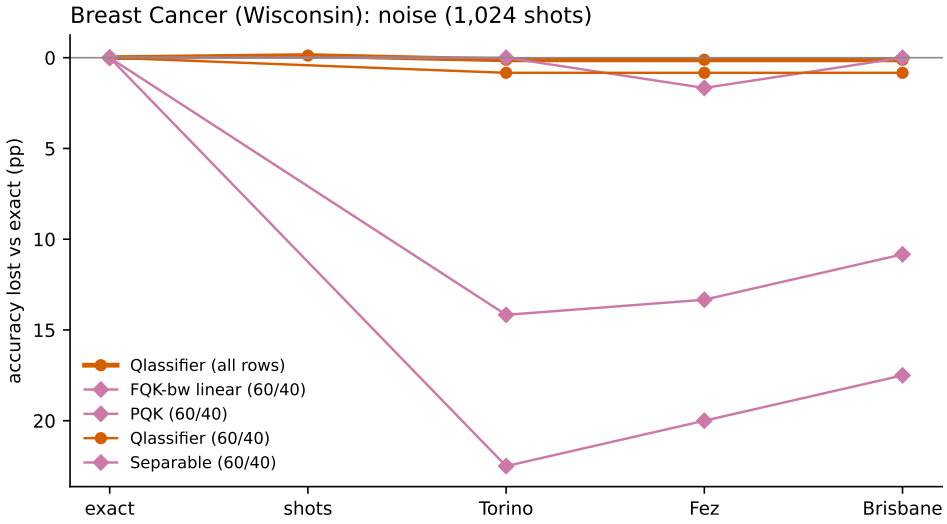


FIG. 4. Accuracy lost from exact simulation to shot noise and three IBM device noise models, Breast Cancer. Thick line: Qlassifier on all rows (Part A). Thin lines: the Part B subsets.

TABLE VI. Every real-device run on ibm_fez: the pre-registered seed 0 and the later seeds. Predicted = FakeFez device model (mean ± sd over shot-noise draws); Δ = hardware − predicted; kernel r = correlation of the measured test kernel with the exact one.

Seed	Job	QPU s	Exact	Predicted	Hardware	Δ (pp)	Within 2 pp	Kernel r
0	`dbesrqnpc3tc739mb92g`	82	0.875	0.880 ± 0.010	0.900	+2.0	yes	0.999
1	`dbeutp8m94ac738silb0`	82	0.975	0.975 ± 0.000	0.975	0.0	yes	0.998
2	`dbeuv97pc3tc739mf740`	82	0.875	0.874 ± 0.010	0.875	+0.1	yes	0.999
3	`dbev0anpc3tc739mf97g`	82	0.975	0.975 ± 0.000	0.975	0.0	yes	0.999
4	`dbev157sstms73fcr4s0`	82	0.950	0.954 ± 0.009	0.950	−0.4	yes	0.999

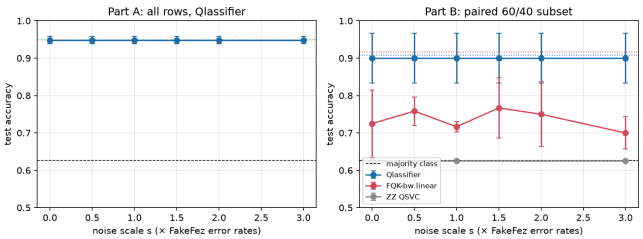


FIG. 5. Accuracy vs noise scale s (FakeFez error rates $\times s$), Breast Cancer paired subsets, 1,024 shots. Dashed: exact simulation.

TABLE VII. Physical qubits of the ibm_fez run and their calibration at submission (2026-10-25).

Physical qubit	T_1 (μs)	T_2 (μs)	$\sqrt{X}$ error (10^{-4})	Readout error (%)
4	117	104	3.0	0.38
2	148	128	1.4	0.45
130	158	153	4.1	0.28
50	108	94	4.1	0.23

I. Cost and time

On Breast Cancer, Qlassifier needs 1,707 circuits to train and predict. The bandwidth-tuned fidelity-kernel pipeline needs 2,423,299, mostly for its on-device bandwidth search: 1,420 times more (BT-7; Table IX, Fig. 6). Counting only its final Gram matrices, without the search, the pipeline needs 147,061 circuits, 86 times Qlassifier's; both readings hold. The projected kernel matches Qlassifier's circuit count but needs two-qubit gates. The full Breast Cancer leaderboard (10 seeds, 12 models) runs in 56.2 s on an Apple M4 laptop with 10 cores (BT-10). There, tuning Qlassifier over its 30 configurations takes a median of 0.38 s per split, and the final fit 0.016 s.

J. Targets

Table X lists the ten targets as the library evaluates them: 9 are met and 1 is report-only (BT-6). BT-6 (beat every tuned classical model with a CI above zero) is report-only: it was missed under the strict protocol before this run (Table IV), and by our pre-registered rule no later run can upgrade it. BT-8 is met on the core datasets (qubits $\leq \lceil d/2 \rceil$, no two-qubit gates) and on the

TABLE VIII. Three more UCI datasets under the same protocol (balanced accuracy, mean ± sd over 10 seeds; corrected CIs).

Dataset	Rows	Qlassifier BA	ZZ QSVC BA	Δ vs ZZ [CI] (wins)	Δ vs best classical [CI]
Spambase	4,601	0.911 ± 0.006	0.899	+1.2 [+0.8, +1.6] (10/10)	−0.8 [−2.0, +0.4] vs Gradient boosting
Banknote	1,372	1.000 ± 0.001	0.955	+4.4 [+2.7, +6.2] (10/10)	0.0 [−0.2, +0.2] vs MLP
Haberman	306	0.564 ± 0.031	0.546	+1.9 [−4.1, +7.9] (6/10)	−1.8 [−8.1, +4.4] vs MLP

TABLE IX. Breast Cancer cost study: qubits, two-qubit gates and depth per circuit after transpilation to IBM FakeTorino, FakeFez and FakeBrisbane, and circuits to train and predict ($N_{\text{train}} = 398$, $N_{\text{test}} = 171$). Fidelity kernels need one circuit per pair of rows; models whose hyperparameters change the circuit also pay for the on-device search.

		2Q gates	Depth	Circuits		
Model	Qubits	Torino / Fez / Brisbane	Torino	train + predict	tuning on device	total
PQK	9	72 / 72 / 72	165	1,707	0	1,707
Qlassifier	4	0 / 0 / 0	6	1,707	0	1,707
ZZ QSVC	8	393 / 428 / 394	623	147,061	0	147,061
FQK-bw full	8	393 / 428 / 394	623	147,061	2,276,238	2,423,299
FQK-bw linear	8	54 / 54 / 54	133	147,061	2,276,238	2,423,299
Separable	8	0 / 0 / 0	6	147,061	2,276,238	2,423,299

protein angles, where geometry mode uses 10 qubits for 10 angles: the $\lceil d/2 \rceil$ bound applies to the core encoding, and geometry mode is judged by one qubit per circle or direction (Sec. VI).

V. RELATED WORK AND NOVELTY

Known parts. Dense angle encoding [11] and trigonometric product feature maps [12]; projected quantum kernels [13]; product and separable kernels as strong baselines [9, 24]; the role of the bandwidth [3–5, 8]; the Fourier view of angle encodings [25, 26]; symmetry-aware and geometric models [7, 27–29]; stereographic Bloch embeddings [30]; directional statistics and torus methods for dihedral angles [15, 16, 31, 32]; noise rescaling of quantum kernels [6, 33] and Pauli decompositions of fidelity kernels [34]; and structure-aware encodings that beat tuned RBF on synthetic or simulated data [35, 36].

What is new here. (i) The design rule: circuit angles fixed by the data scale or by the declared geometry, bandwidth and margin tuned only after measurement. It costs zero tuning circuits and keeps the device noise loss small. (ii) Geometry mode: one qubit per circle or direction, no aliasing, with the exact classical twin reported next to it. (iii) The hardware protocol: three local-Pauli settings per sample ($O(N)$ circuits) with per-qubit unit renormalisation. (iv) A benchmark of all of this against every challenge baseline on real data, with leakage-safe splits, corrected statistics and the negative results kept (Sec. VI). We found no prior applied QML on protein dihedral angles with angle-native encodings.

VI. LIMITATIONS

- *No quantum advantage.* Qlassifier prepares product states, which a classical computer simulates exactly; the library's default mode does just that. In geometry mode the classical twin reproduces the kernel by construction. Our results are measured gains over quantum baselines and ties with classical ones.

- *BT-6 missed.* We did not beat every tuned classical model on any dataset under a leakage-safe protocol. A residue-level split showed +1.4 pp (CI [+0.3, +2.4]) over MLP, but it shares protein chains between training and test. Under chain-grouped splits the margin shrank to +0.7 pp (CI [−0.6, +2.1]), and two pre-registered tests on 1,773 chains gave +0.25 pp (97.5% CI [−1.13, +1.62]) and +0.81 pp (97.5% CI [−0.61, +2.23]).

- *A loss: global earthquakes.* On 3,729 earthquakes labelled by depth from their latitude and longitude (USGS), geometry mode lost to Random forest by −6.6 pp (CI [−8.5, −4.8]), although it beat RBF-SVM by +5.6 pp. A wider kernel grid closed part of the gap (−2.1 pp, CI [−5.2, +1.1]). Trees adapt to the uneven density of subduction zones; one global kernel width does not. The narrow kernels this task needs also need far more than 1,024 shots.

- *Limited hardware evidence.* Beyond the device models we have 5 real runs (Sec. IV G) on one device under one day's calibration, 40 test rows each; only the first was pre-registered. The device models have no crosstalk, drift or leakage, no idle noise and symmetric readout errors; the per-qubit sim-

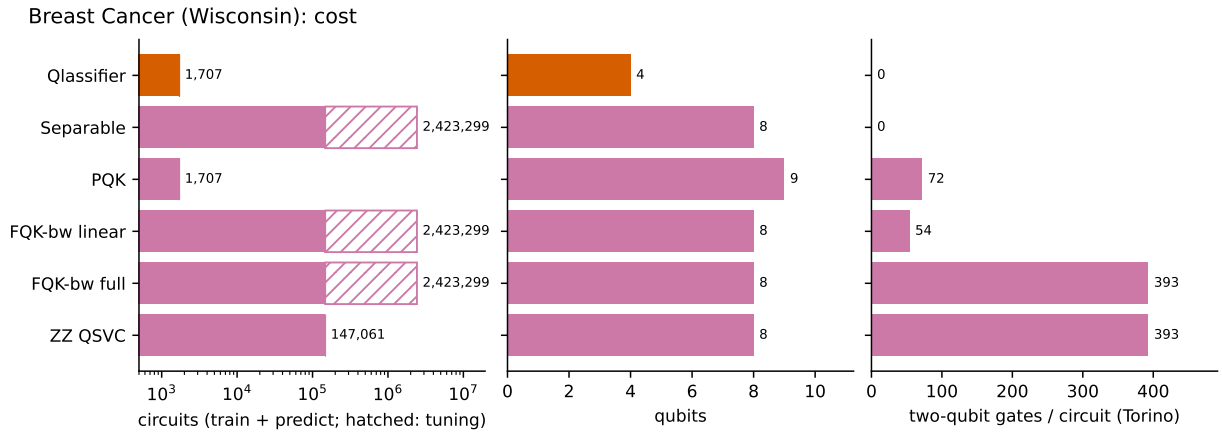


FIG. 6. Breast Cancer cost per quantum model: circuits to train and predict (log scale; hatched: on-device tuning), qubits, and two-qubit gates per circuit on FakeTorino.

ulation is exact only under these assumptions. At 1,024 shots the simulated study cannot rank the three devices (Sec. IV).

- *The fidelity kernel's loss is mostly shot noise.* Its 17.5–22.5 pp loss under device models is 19.2 pp with device noise off (Sec. IV F); Qlassifier's advantage over it comes from shot efficiency: three settings per sample instead of one fidelity per pair.

- *The ZZ margin varies.* On Spambase and Banknote the lead over the ZZ baseline is +1.2 and +4.4 pp, not tens of points, and on Haberman every model ties (Sec. IV H).

- *Not in the library yet.* The hardware run, the noise sweep and the three extra datasets are research results with scripts in the repository; the released library has no hardware backend or noise-sweep API (it does benchmark a user's CSV file under the same protocol). They are planned for a later release.

- *More qubits in geometry mode.* Geometry mode uses one qubit per circle, above the $\lceil d/2 \rceil$ of the core encoding (10 qubits for the 10 protein angles), because two circles cannot share a qubit (Sec. II D). The hardware target counts geometry mode by this rule; the circuits stay free of two-qubit gates, so the device cost per circuit is unchanged.

- *Small data, wide intervals.* The UCI test sets hold 63 to 171 rows. With 10 seeds, the corrected CIs are wide.

- *Scope.* Binary classification of tabular data with up to 10,000 training rows (the Gram matrix is $N \times N$).

VII. REPRODUCIBILITY

Install the library from the repository (github.com/szymon6/qlassifier) with `pip install .` (Python 3.12), then run

```
qlassifier benchmark --noise
```

to reproduce every leaderboard, noise and cost number of this paper. The saved results come from two runs on one Apple M4 laptop (10 cores): the UCI datasets with 10 workers (about 14 min, commit 335e996) and the protein angles with 3 workers (about 28 min, commit d1d218c), because ten parallel 10-qubit noise cells exhausted the laptop's memory. The worker count changes wall time only: every task is seeded by dataset, seed and model, and the two versions produce bit-identical device samples (`benchmarks/README.md`). Before the run, the engine reproduced 480 of 480 rows of the method search's reference leaderboard with 0 mismatches (chosen configuration and every metric within 10^{-9}; `benchmarks/golden_e01.txt`). Across platforms, BLAS rounding can move a ROC-AUC by one resolution step and the MLP by more. Every result file records the versions (Qlassifier 0.1.0, Qiskit 2.6.1, Qiskit Aer 0.18.0, scikit-learn 1.9.1, NumPy 2.5.4) and the protocol fingerprint (`33a530ad036d`). Results depend only on dataset, seed, model and versions, not on scheduling. The saved results, per-seed tables and plots are in `benchmarks/`. The script `scripts/paper_tables.py` writes every number of this paper from them, and the CI fails when they drift apart. The method-search experiments behind Table IV are in `experiments/classifier/`. The results (Secs. IV F, IV G, IV H) are in `experiments/hardware/` with their scripts and raw outputs; the hardware record is `e07-hardware-point/run-seed0/record.json`, and the same script reproduces the numbers of those sections. The command `qlassifier ui` opens a local dashboard with the same leaderboards, noise and cost views.

TABLE X. Benchmark targets BT-1 to BT-10 as evaluated by `qlassifier benchmark` (`targets.json`).

Target	Pass rule	Measured	Status
BT-1	Breast Cancer: mean accuracy Qlassifier − ZZ QSVC ≥ 20 pp	0.956 vs 0.677 → +27.9 pp	met
BT-2	mean BA Qlassifier > ZZ QSVC on all 5 datasets; paired win rate ≥ 0.8 on each	breast_cancer +37.6 pp (10/10); ionosphere +22.6 pp (10/10); sonar +27.9 pp (10/10); heart +28.3 pp (10/10); protein_angles +35.5 pp (10/10)	met
BT-3	mean BA ≥ published − 1 pp on ≥ 3 of 4 UCI sets (0.968 / 0.911 / 0.813 / 0.804)	breast_cancer 0.948 (−2.0 pp); ionosphere 0.925 (+1.4 pp); sonar 0.824 (+1.1 pp); heart 0.816 (+1.2 pp) → 3/4	met
BT-4	mean BA ≥ published on ≥ 2 of 4 UCI sets	breast_cancer 0.948 (−2.0 pp); ionosphere 0.925 (+1.4 pp); sonar 0.824 (+1.1 pp); heart 0.816 (+1.2 pp) → 3/4	met
BT-5	mean BA over the 4 UCI sets: Qlassifier highest among the in-harness quantum models	qlassifier 0.878, separable 0.870, fqk_bandwidth_full 0.864, pqk 0.862, fqk_bandwidth_linear 0.856, zz_qsvc 0.587	met
BT-6	≥ 1 dataset: mean BA above every tuned classical model and corrected 95% CI of Δ vs best classical > 0 (stretch; report-only)	breast_cancer vs mlp: −1.3 [−3.1, +0.5] pp (tie); ionosphere vs rbf_svm: −0.2 [−3.1, +2.7] pp (tie); sonar vs mlp: +0.5 [−8.5, +9.5] pp (tie); heart vs logistic_regression: −1.0 [−3.1, +1.2] pp (tie); protein_angles vs gradient_boosting: +1.0 [−0.2, +2.2] pp (tie)	report-only
BT-7	Breast Cancer: ≥ 100× fewer circuits than the bandwidth-tuned fidelity-kernel pipeline (incl. its search)	1,707 vs 2,423,299 → 1,420×	met
BT-8	per circuit on FakeTorino / FakeFez / FakeBrisbane: core encoding qubits ≤ ceil(d/2); geometry mode: 1 qubit per circle or (lat, lon) pair + ceil(d_plain/2); 2Q gates ≤ n − 1	breast_cancer: 4 qubits (core, ceil(8/2) = 4), 0 two-qubit gates; ionosphere: 4 qubits (core, ceil(8/2) = 4), 0 two-qubit gates; sonar: 4 qubits (core, ceil(8/2) = 4), 0 two-qubit gates; heart: 4 qubits (core, ceil(8/2) = 4), 0 two-qubit gates; protein_angles: 10 qubits (geometry limit 10), 0 two-qubit gates	met
BT-9	Breast Cancer, 1,024 shots: mean accuracy drop ≤ 2 pp on each device (part A); < FQK-bw drop (part B)	part A drop 0.12 / 0.12 / 0.12 pp (Torino / Fez / Brisbane); part B Qlassifier 0.8 / 0.8 / 0.8 vs FQK-bw linear 22.5 / 20.0 / 17.5 pp	met
BT-10	`qlassifier benchmark breast_cancer`, defaults, exact, Apple M-class laptop: wall ≤ 120 s	56.2 s on Apple M4 (10 cores, macOS-27.0-arm64-arm-64bit; 1-min load 1.95 at start)	met

VIII. CONCLUSION

A quantum kernel classifier can be good by default and cheap on hardware if it keeps its circuits fixed and tunes after the measurement. Against the standard ZZ baseline, Qlassifier gains 22.6 to 37.6 pp BA on the UCI datasets; it ties tuned classical models, has the highest mean BA of the quantum models in our harness, and loses at most 0.19 pp under IBM device noise models, with few qubits and no two-qubit gates. 5 runs on an IBM Heron processor returned the accuracy the device model predicted, for 410 QPU seconds in all. Where the data has geometry, geometry mode encodes it directly, but the gain comes from the geometric prior, as the classical twin shows. The benchmark, including its ties and losses, ships with the library for reuse.

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